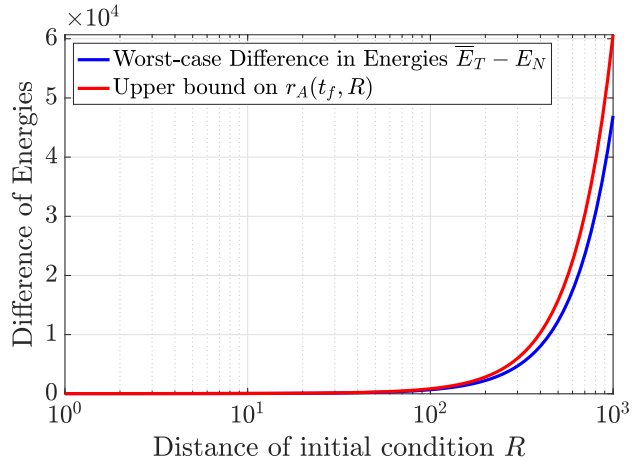

FIGURE 1. Illustration of energetic resilience metric as a function of distance of the initial condition from the origin for the linear driftless model of an underwater robot. The actual difference in energies $E_T - E_N$ does not vary significantly with different choices of uncontrolled inputs u_{uc}, and is hence not shown here.

We remark that the model (49) is a minor modification of the model considered in [20]. Since that model heavily biases the effect of u_1 on x, this metric may be relatively less useful in that case in comparison to the model in (49). The heavier weight contributes to poor conditioning of the matrices $B^{\dagger T}B^{\dagger}$ and $B_c^{\dagger T}B_c^{\dagger}$, resulting in a more conservative bound in (23) than in the example presented here.

6.2. **Example 2: Linear Model.** In this example, we consider the ADMIRE fighter jet model [44], a popular benchmark for control applications [15, 45]. The linearized dynamics $\dot{x} = Ax + Bu$ for the subsystem associated with control actions were established in [44], where

$$A = \begin{bmatrix} -0.9967 & 0 & 0.6176 \\ 0 & -0.5057 & 0 \\ -0.0939 & 0 & -0.2127 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & -4.2423 & 4.2423 & 1.4871 \\ 1.6532 & -1.2735 & -1.2735 & 0.0024 \\ 0 & -0.2805 & 0.2805 & -0.8823 \end{bmatrix}. \tag{50}$$

The state $x = [p, q, r]^T$ consists of the roll, pitch and yaw rates measured in rad/s. The control input $u = [u_c, u_{re}, u_{le}, u_r]^T$ consists of deflections of the canard wings, right and left elevons and the rudder, measured in radians. Note that these inputs correspond to deflections from their equilibrium values in the linearization. We illustrate the use of the energetic resilience metric by calculating approximate control energies based only on the deflections.

Under the dynamics $\dot{x} = Ax + Bu$, where A and B are given in (50), we have $D_f = \|A\|_\infty = 1.6143$ and $D_g = 0$. We consider a loss of control authority over the canard wing so that B_{uc} consists of the first column of B in (50). It was shown in [15] that the system is resilient to the loss of this control input. We fix $t_f = 1$ and vary the distance R of the initial condition x_0 from the target state x_{tg}, chosen to be $[0\ 0\ 0]^T$. We remark that while we do not directly check whether achieving this target state is always possible, tools for continuous reachability analysis [46] can be used to verify this.

Fig. 2 compares the upper bound on $r_A(t_f, R)$ in (48) under the linear model for this example with the worst-case difference in energies $\overline{E}_T - E_N$ computed using the approximations (32) and (39). These quantities are plotted as a function of R, with $\|\tilde{x}\|_2 \leq R$. The thinner dashed lines in this figure show the difference between the approximate total and nominal energies $E_T - E_N$ for various uncontrolled inputs u_{uc}, computed using (32), (38) and the definition (10). We test uncontrolled inputs ranging sinusoids of varying frequencies, constants of different amplitudes and exponential decays.

It is clear that (48) bounds the difference in energies $E_T - E_N$, even under the worst-case uncontrolled input where we have the difference $\overline{E}_T - E_N$. As R increases, we note that the bound on $r_A(t_f, R)$ increases faster than the difference in energies, presumably because $\overline{E}_T$, E_T and E_N all increase with R. This behavior

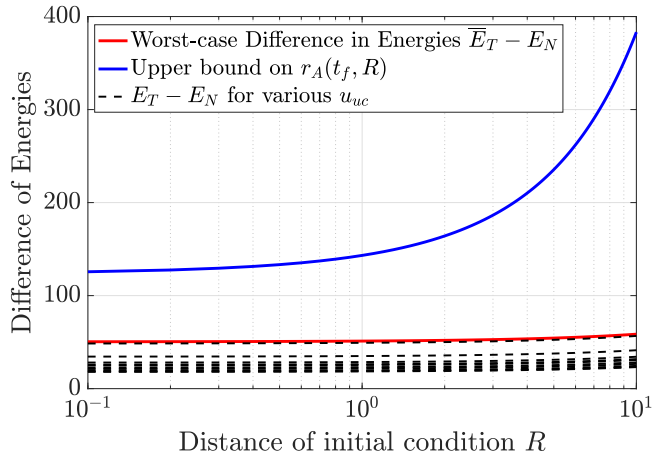

FIGURE 2. Difference between nominal and malfunctioning energies, compared to the upper bound on the metric $r_A(t_f, R)$, for a linear model of a fighter jet. The thinner dashed lines plot $E_T - E_N$ for a variety of uncontrolled inputs u_{uc}.

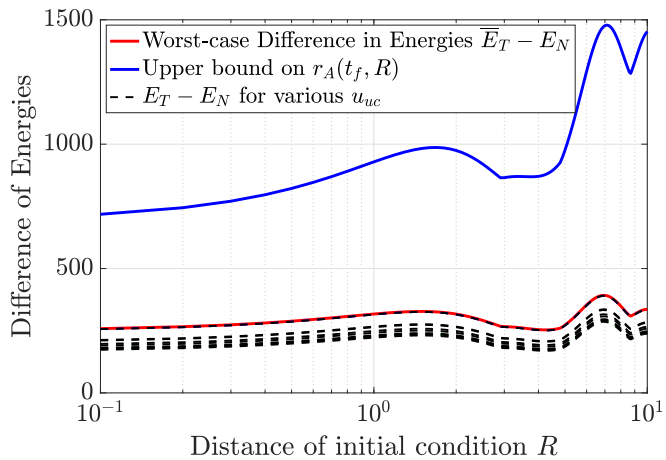

FIGURE 3. Difference between nominal and malfunctioning energies, compared to the upper bound on the metric $r_A(t_f, R)$, for a nonlinear wind-affected model. The thinner dashed lines plot $E_T - E_N$ for a variety of uncontrolled inputs u_{uc}.

contributes to smaller errors between the worst-case difference $\overline{E}_T - E_N$ and the bound on $r_A(t_f, R)$ when R is smaller. We further note that these quantities are of the same order of magnitude for the range of R considered, even when the relative error between them is large. Fig. 2 is thus an example of how the metric $r_A(t_f, R)$ can be used to estimate the maximal additional energy required to achieve a target state for a linear system.

6.3. **Example 3: Wind-induced Nonlinear Model.** In this example, we consider the dynamics of the ADMIRE fighter jet model subjected to wind effects. The dynamic model can then be written as $\dot{x} = Ax + Bu + f_w(x)$, where A and B are given in (50) and $f_w(x)$ is a nonlinear function of the state, modeling the effect of wind on the fighter jet model. We assume $f_w(x)$ consists of sinusoidal and constant expressions in each component as modeled in [47], and that its Lipschitz constant is no more than 1, so that $D_f = \|A\|_\infty + 1 = 2.6143$. In this particular instance, we choose $f_w(x) = \frac{1}{2}\left[\sin(p)\cos^2(p),\ -\sin(2q),\ 1\right]^T$. We note that when $f_w(x)$ is unknown, we can test the applicability of the metric by searching over functions with Lipschitz constants smaller than 1. As the dynamics are still linear in the input, we have $D_g = 0$. As before, we consider a loss of control authority over the canard.

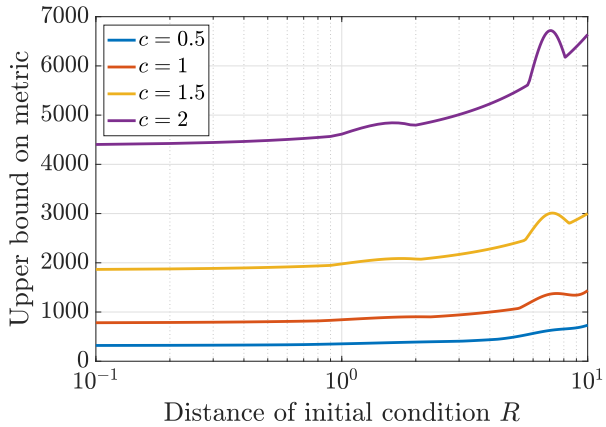

FIGURE 4. Sensitivity of metric to different amplitudes of wind disturbance.

In Fig. 3, we compare the upper bound on $r_A(t_f, R)$ in (48) to the worst-case difference in energies $\overline{E}_T - E_N$, where we use the approximations (32) and (39) to compute these quantities. We plot these quantities as a function of the distance R of the initial condition x_0 from the target state $x_{tg} = [0\ 0\ 0]^T$, and fix the final time as $t_f = 1$. As before, tools for reachability analysis can be used to verify whether the target state can be achieved under these settings. The thinner dashed lines denote the difference in energies $E_T - E_N$ for different uncontrolled inputs u_{uc}, computed using the approximations (32), (38) and the definition (10). As before, the uncontrolled inputs we test consist of sinusoids of varying frequencies, exponential decays and constant inputs of different amplitudes. It is clear that even under the worst-case uncontrolled input, the difference in energies $\overline{E}_T - E_N$ is bounded by (48). As expected, the actual differences in energies $E_T - E_N$ for different uncontrolled inputs is also bounded by (48).

While the relative error between the worst-case difference $\overline{E}_T - E_N$ and the bound on $r_A(t_f, R)$ is large, we note that these quantities are of the same order of magnitude. This error is also smaller when x_0 is closer to x_{tg}, i.e., when R is small. In this case, the bound on $r_A(t_f, R)$ is less than 3 times larger than the worst-case difference in energies. We also note that the results for the nonlinear model in Fig. 3 show an improvement over the results for the linear model in Fig. 2. This is due to the nonlinear function $f_w(x)$ that models wind effects. The energy required in the control signal can decrease when this function acts in the same direction that the system seeks to move towards.

In Fig. 4, we illustrate the sensitivity of the metric to different amplitudes of wind disturbances. In particular, we test disturbances of the form $f_w(x) = \frac{c}{2}\left[\sin(p)\cos^2(p),\ -\sin(2q),\ 1\right]^T$, where c is a varying parameter signifying the amplitude of disturbances. Note that $c = 1$ is the case considered in Fig. 3. Clearly, the upper bound on the metric is sensitive to the choice of c, and for larger values of c, may exhibit more conservative behavior.

We remind the reader that the expressions plotted in Figs. 2 and 3 are based on approximations made in (32) and (38), resulting in approximations for $\overline{E}_T$ (39) and $r_A(t_f, R)$ (48). Under these approximations, we note that $r_A(t_f, R)$ is a useful quantity that estimates the increased energy required without significant conservatism, despite our knowledge of only its approximate upper bound.

7. CONCLUSIONS

In this article, we introduce a metric to bound the maximal additional energy used by a nonlinear dynamical system in achieving a fixed-time reachability task, when it loses authority over a subset of its actuators. This maximal energy is compared to the nominal energy expended when no authority is lost over the inputs. We first consider the special case of linear driftless systems and recall the energies used by the control signal in both the nominal and malfunctioning cases. This analysis is used to derive a bound on the metric to quantify the maximal additional energy used in the malfunctioning case, assuming control authority is